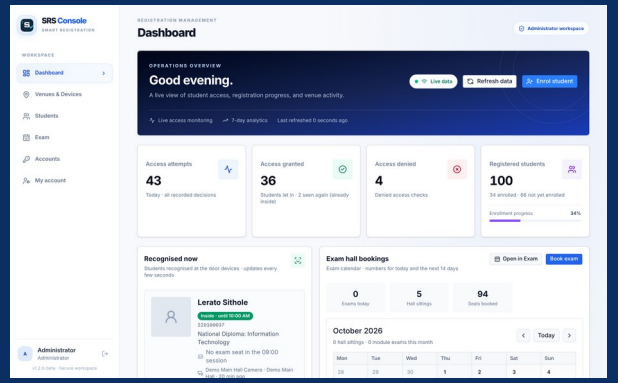
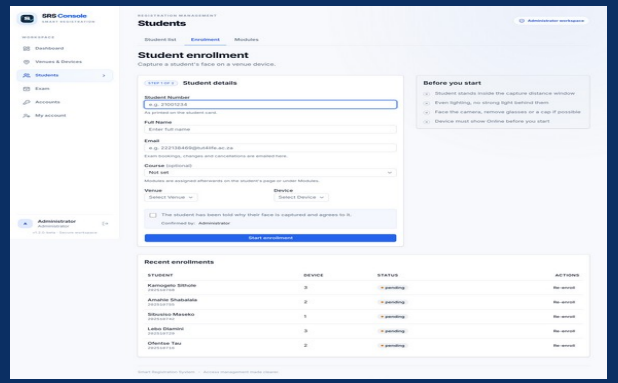
Tshwane University of Technology

We empower people

Smart Registration System

Face-recognition access control for university exam venues





TC Mance · 222138469 · Project Design 376 (PD2316D)
Lecturers: Mr M. Muwanguzi Ntume and Prof C. Du · 5 November 2026

Checking identity by eye is slow and easy to fool

Slow at the door

Every candidate is compared with a register and a student card

Interrupts writing

Checks are repeated during the exam, disturbing students

Impersonation slips through

A convincing look-alike passes a quick visual check

No reliable record

Paper ticks give no time-stamped audit trail

The aim

Verify every candidate automatically at the venue door

Show each student's seat and session at the door

Keep the invigilator in charge of every refusal

Four objectives from the proposal – all met

Fast, distance-gated recognition

MET

Goal: Face verified within 2 s at 90%+ accuracy

≈1 s hold + 16 ms recognition · 98.6% accuracy

Low FAR and FRR

MET

Goal: Keep both error rates acceptable

FAR 0.03% · FRR 2.8% at threshold 0.40

Log every attempt, live dashboard

MET

Goal: Time, student and decision for each entry

Every decision stored; live console for all venues

Manual override

MET

Goal: Backup when recognition fails

Override button records who and why

Face recognition exists – exam-aware access does not

Commercial products

Facephi · SmartVisitor · Navon · Openitem

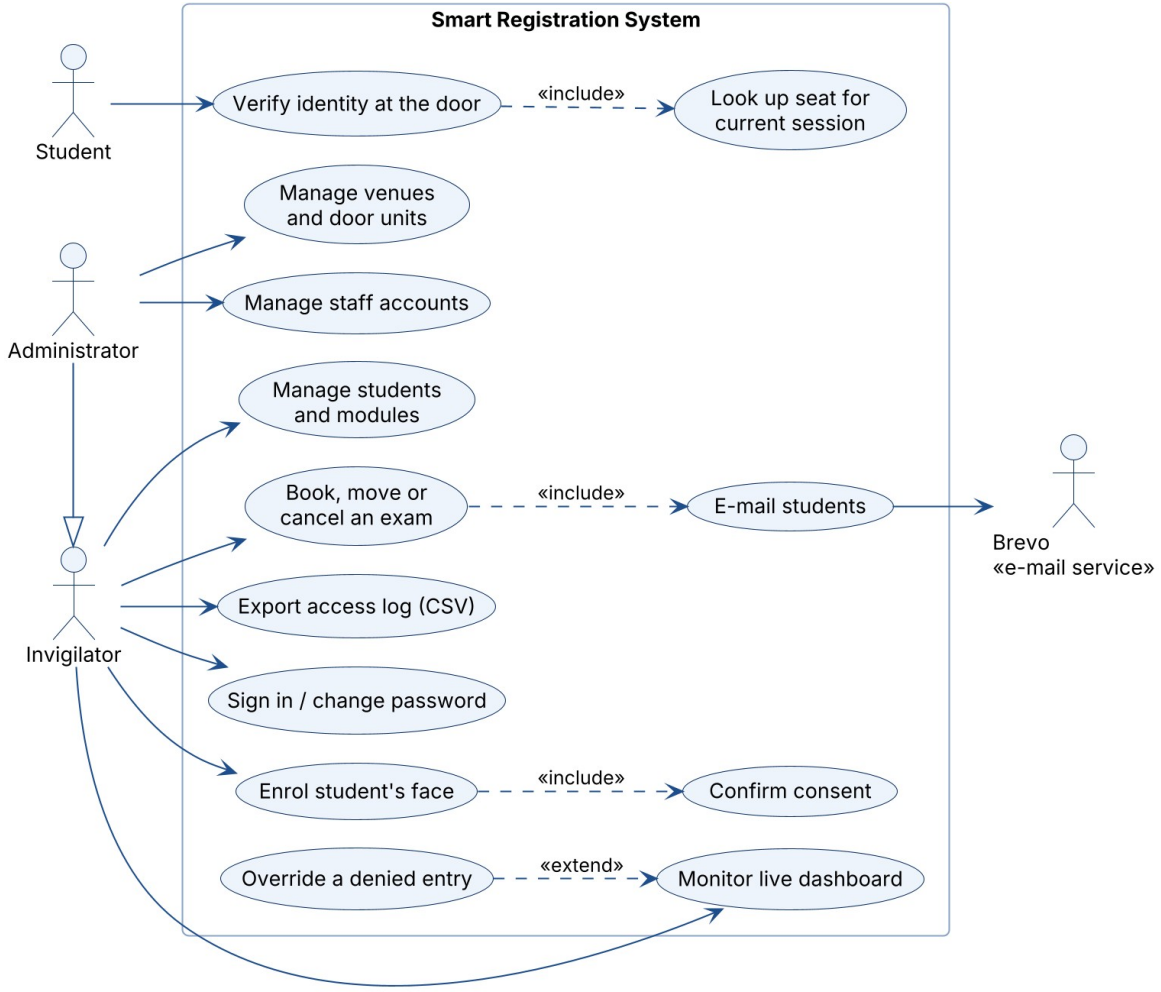
- Offices, estates, events and visitors
- No idea of exam timetables or seats
- Face templates usually kept in the cloud
- Subscription – about R480 per month

Smart Registration System

Built for TUT exam venues

- **Shows each student's seat; flags "No seat" for the invigilator**
- **Once per session; automatic sign-out at 12:00 / 16:00**
- **Face templates stay on the door PC (POPIA)**
- **About R1 225 of hardware, free hosting tiers**

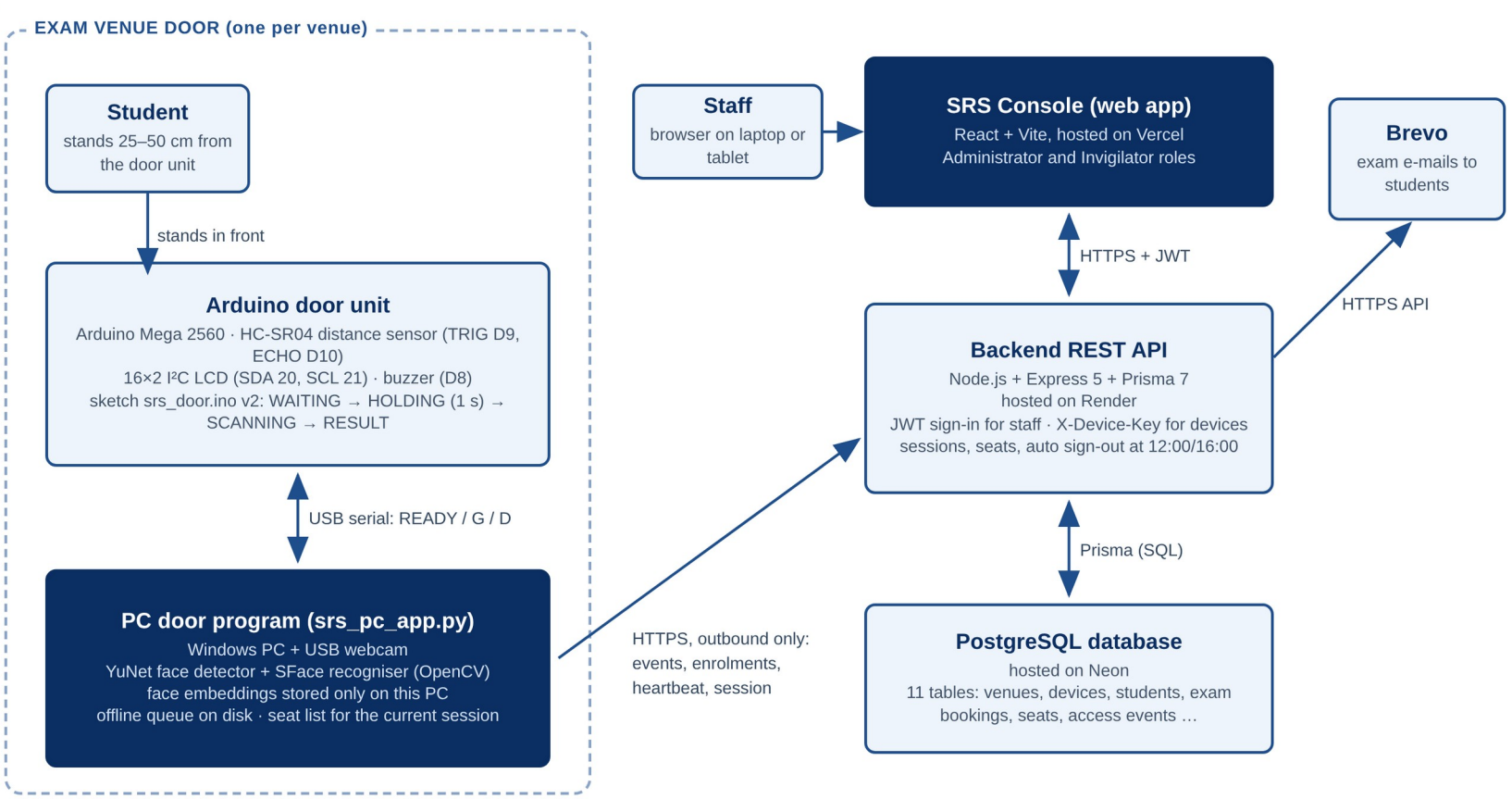
Who uses the system: UML use case diagram



How to read it

- Stick figures are actors: Invigilator, Administrator, Student and the Brevo e-mail service
- Ellipses are what the system lets them do
- Hollow triangle: an Administrator can do everything an Invigilator can, plus venues, devices and accounts
- «include» always happens as part of another use case; «extend» only sometimes (an override)

Four parts, connected only through the backend



Door unit

PC + webcam recognises; Arduino measures distance, shows the result

Backend

Node.js + Express on Render; decides sessions, seats, sign-out

Security

Staff sign in (JWT); doors use a secret device key; outbound HTTPS only

The hardware change: ESP32-CAM to PC + Arduino

Planned

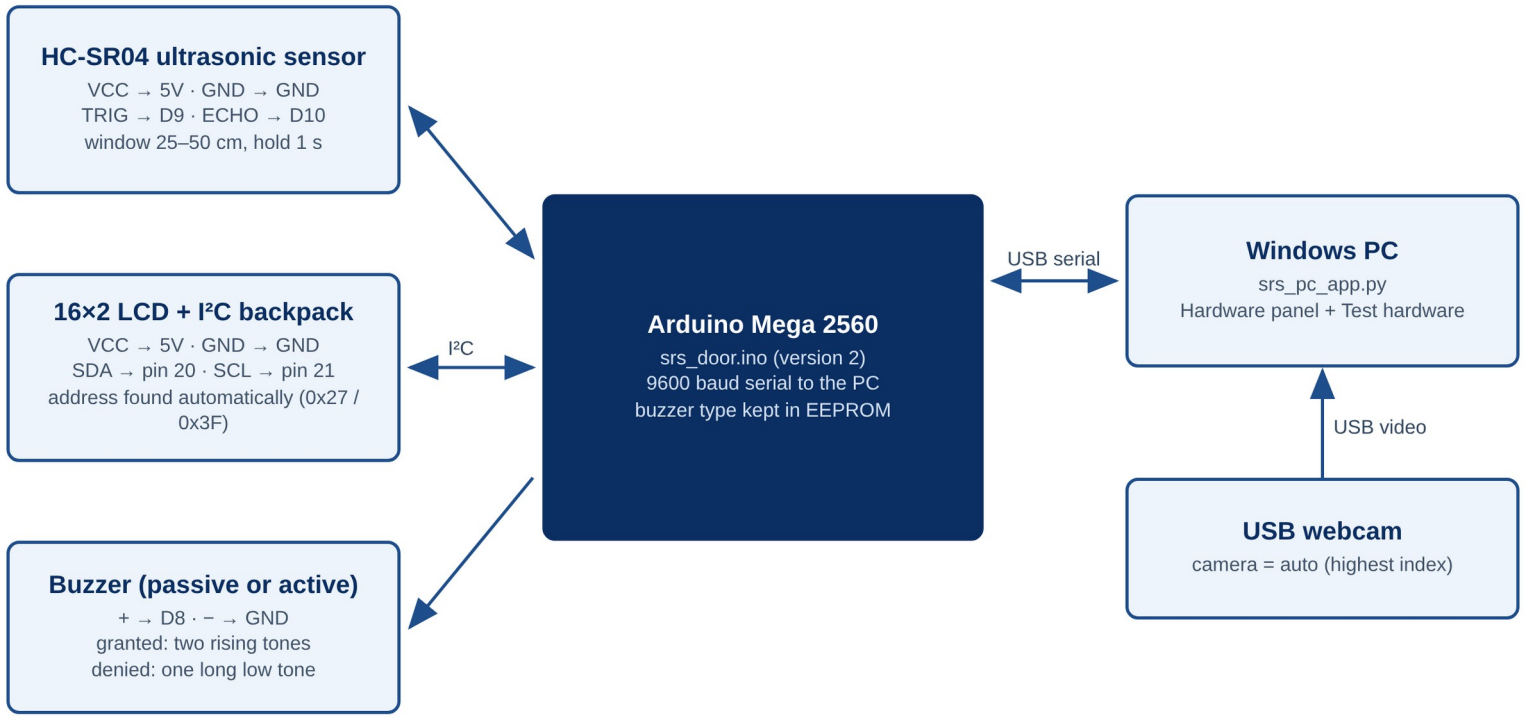
- ESP32-CAM with OV2640 camera
- ESP-WHO recognition on the board
- Battery pack + buck converter
- LED and buzzer only

Final

- **USB webcam on a Windows PC**
- **OpenCV YuNet + SFace recognition**
- **USB power – no batteries**
- **Arduino Mega: distance sensor, LCD, buzzer**

Why: hardware problems with the ESP32-CAM could not be solved in time. The new design removed its memory, power and pin limits – and the new door unit uses the same backend API.

The door unit waits for the student to hold still



25–50 cm

distance window (HC-SR04)

1 s

hold before the picture

G / D / A

answer on LCD + beep

d = 343 m/s × t ÷ 2 → 25–50 cm = 1.45–2.9 ms echo

From face to decision: match first, then session rules

1. Hold

Student holds still 1 s at 25–50 cm

2. Detect

YuNet finds the largest face

3. Match

SFace: 128 numbers, compared with ALL faces, ≥ 0.40

4. Session

Already inside? Session over? Seat looked up

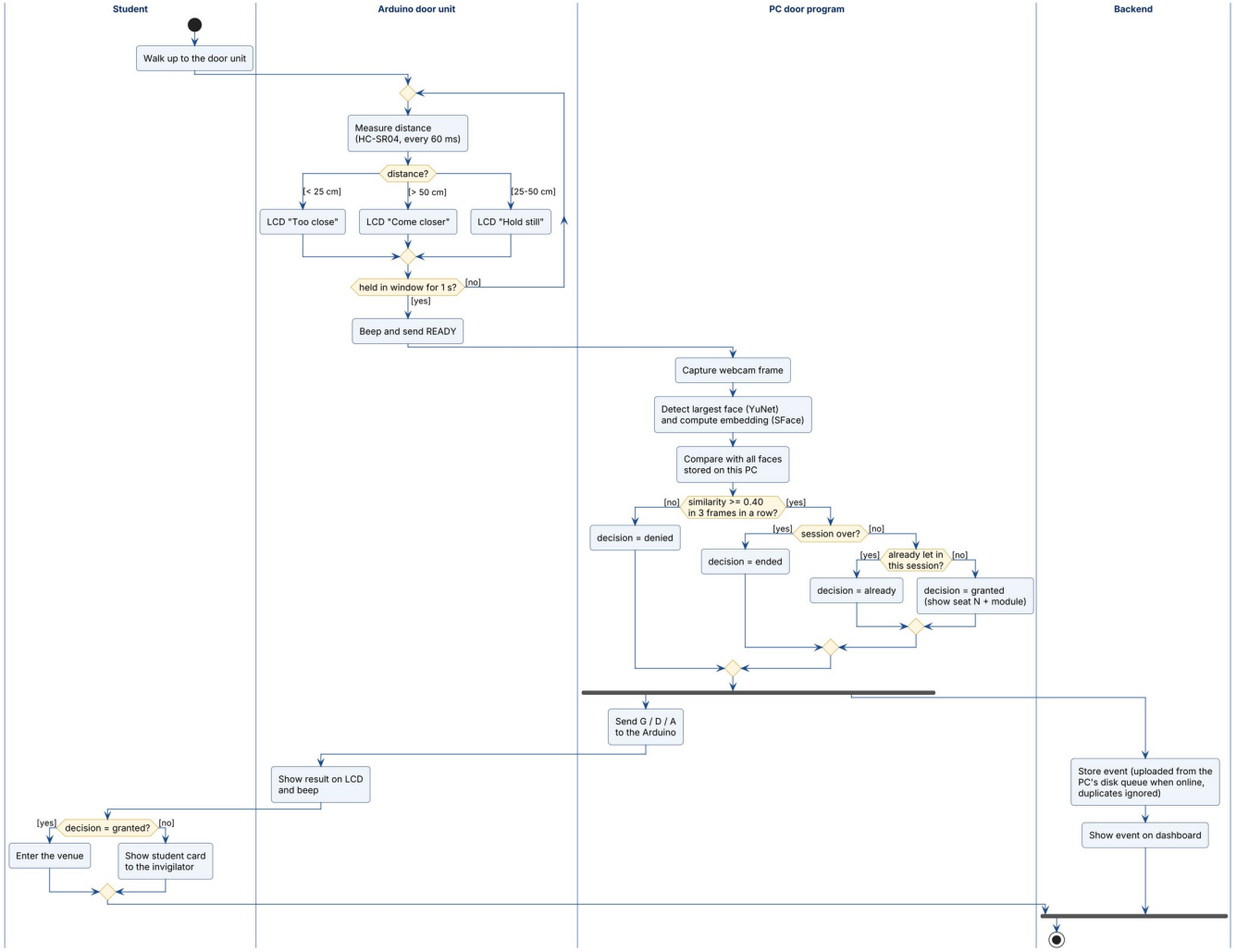
5. Decide

Let in with seat N (or "no seat" flag); unknown face DENIED

Offline? Decisions continue; events are queued on disk and uploaded once, never twice

Face templates stay on the door PC; the backend only gets the result and one enrolment photo

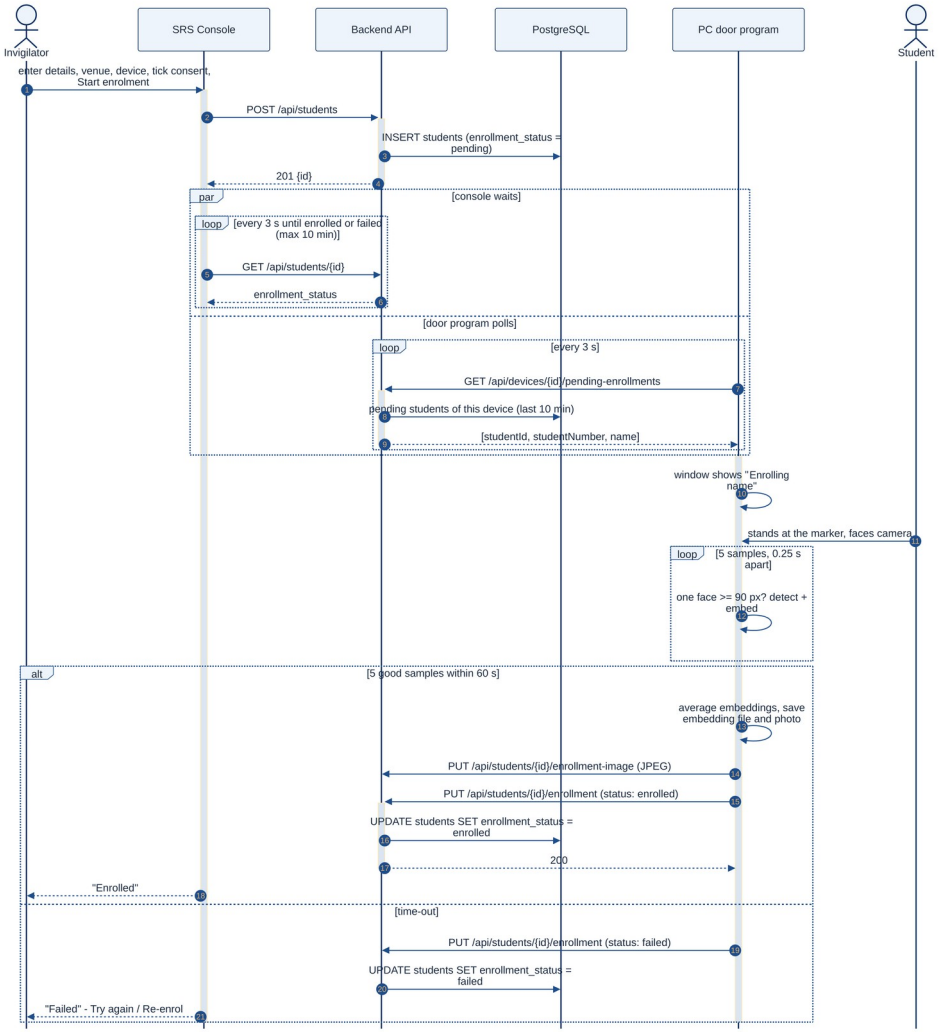
Access verification: UML activity diagram



How to read it

- One swimlane per part: Student, Arduino, PC door program, Backend
- Black dot = start, ringed dot = end
- Diamonds are decisions; the condition for each path is in [square brackets]
- Thick bars: the LCD result and the upload run at the same time (fork), then join

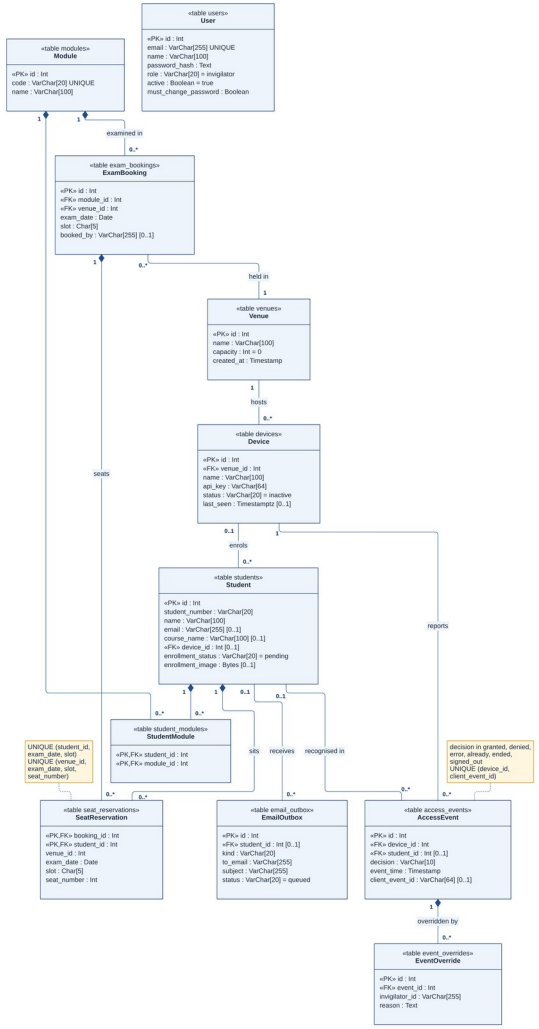
Enrolment: UML sequence diagram



How to read it

- Each participant has a lifeline; time runs downwards
- Solid arrows are requests, dashed arrows are replies
- par: the console and the door program both poll every 3 s
- loop: five face samples; alt: enrolled or time-out after 60 s

Database: UML class diagram (11 tables)



How to read it

- One class per table: columns and data types; «PK» primary key, «FK» foreign key
- Multiplicities on each line: 1, 0..1, 0..* (any number)
- Filled diamond = composition: seats, overrides and module registrations are deleted with their owner
- Venue → Device → Student → AccessEvent is the path of every entry
- User (staff accounts) stands alone: it has no links

SRS Console: one app for invigilators and admins

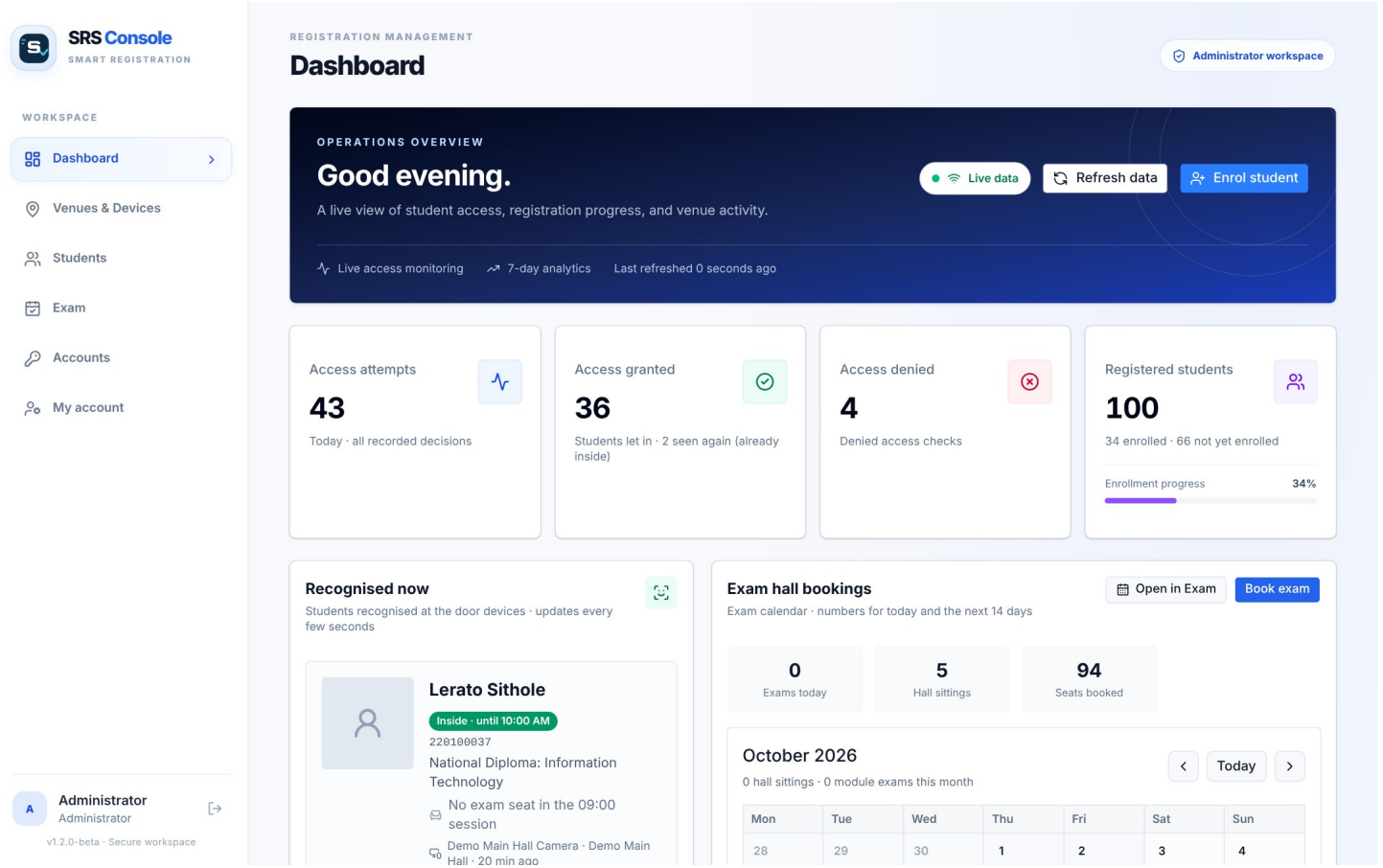
Live dashboard
entries, refusals, recognised now

Enrolment
with consent tick box

Exam bookings
seats, capacity, calendar

E-mails
booked, moved, cancelled

Two roles
admin and invigilator



Demo data shown

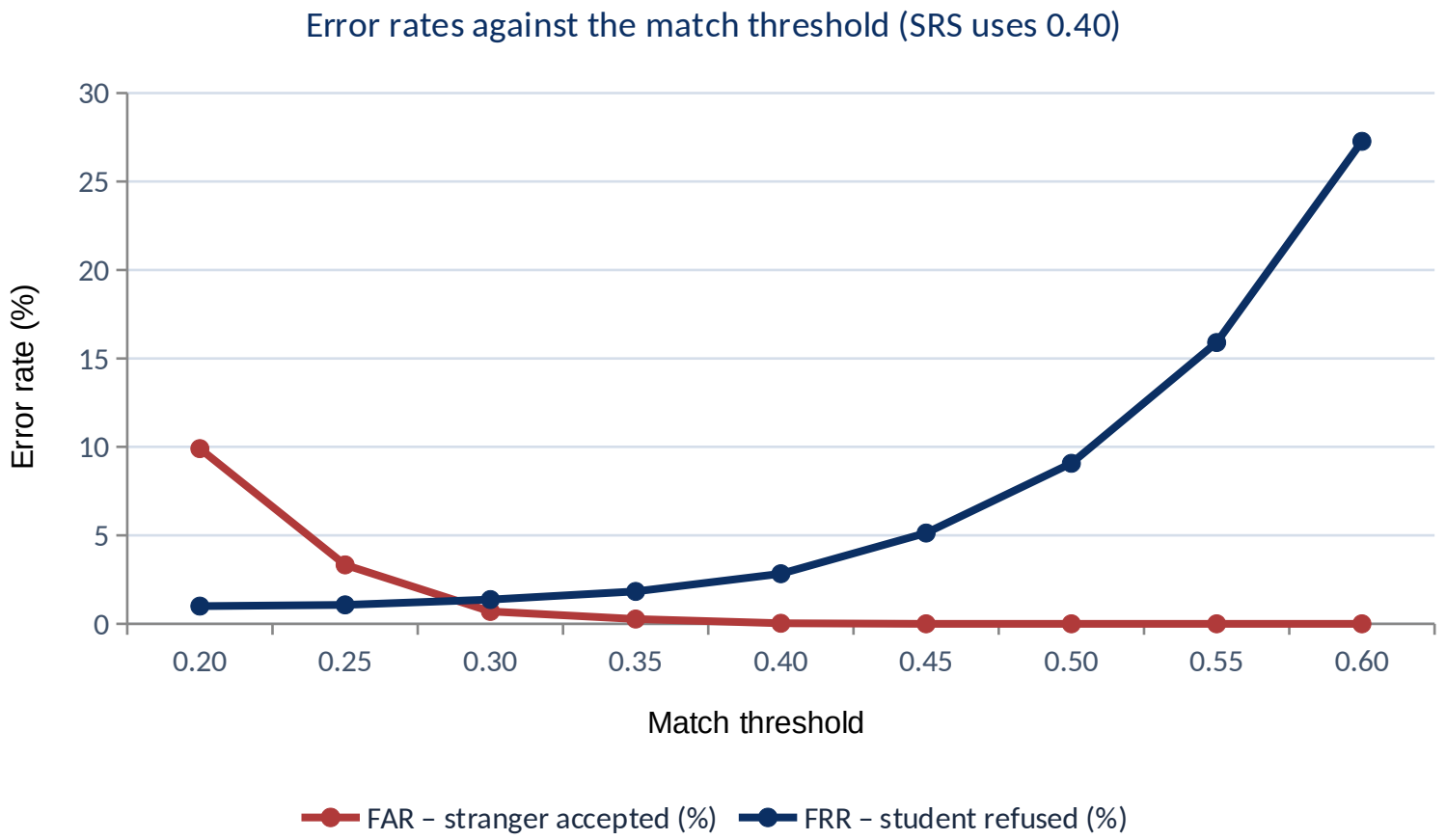
Benchmark: the door code on 13 233 public face photos

98.6%
accuracy (1:1)

0.03%
false accepts

2.8%
false rejects

16 ms
per webcam frame



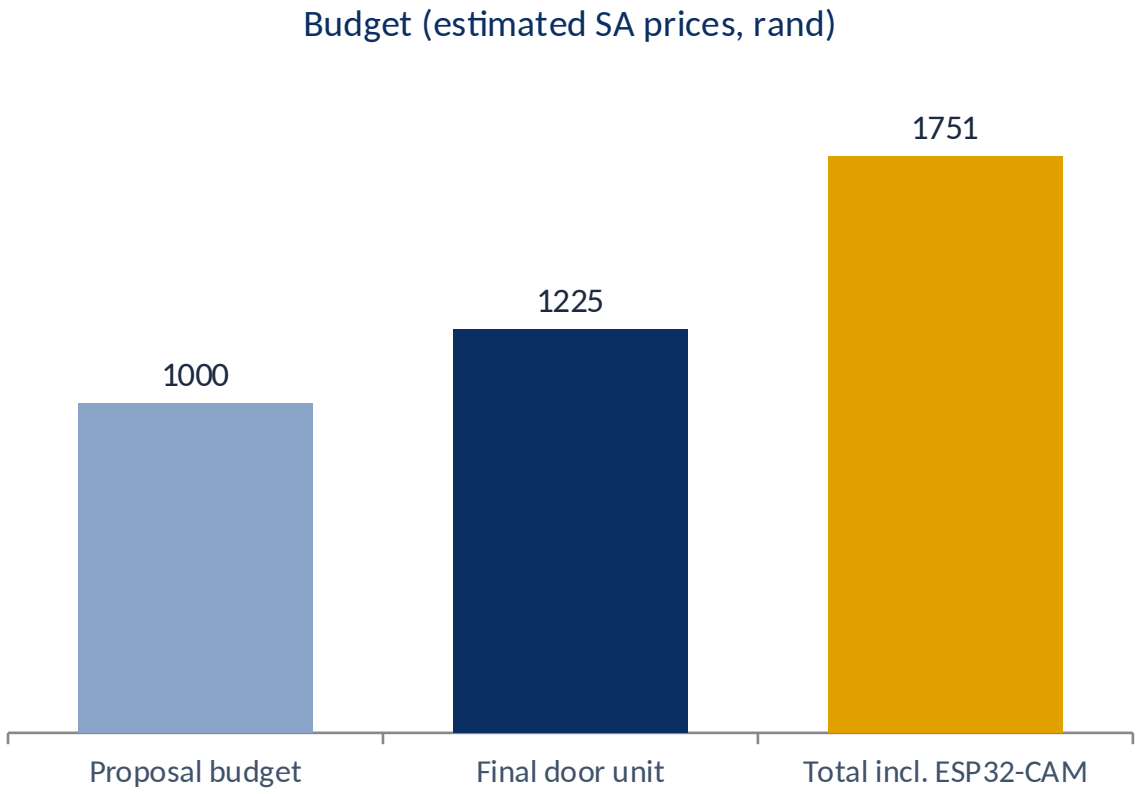
LFW dataset, 3 000 same-person + 3 000 different-person pairs; door simulation with 100 enrolled students: 97.5% correct entries

Tested in code, costed in rand

108 / 108

automated backend tests pass

Area	Tests
Venues, devices, enrolment, events	38
Bookings, seats, capacity	26
Validation, statistics, key replacement	26
Sign-in, roles, lock-out	13
Session end and sign-out	5



Over budget because two sets of hardware were bought after the ESP32-CAM was dropped.

Conclusion and next steps

What was achieved

- All four proposal objectives met
- Recognition tied to exam bookings and seats
- Biometric templates stay on the door unit
- Grew to many venues, e-mails and staff roles

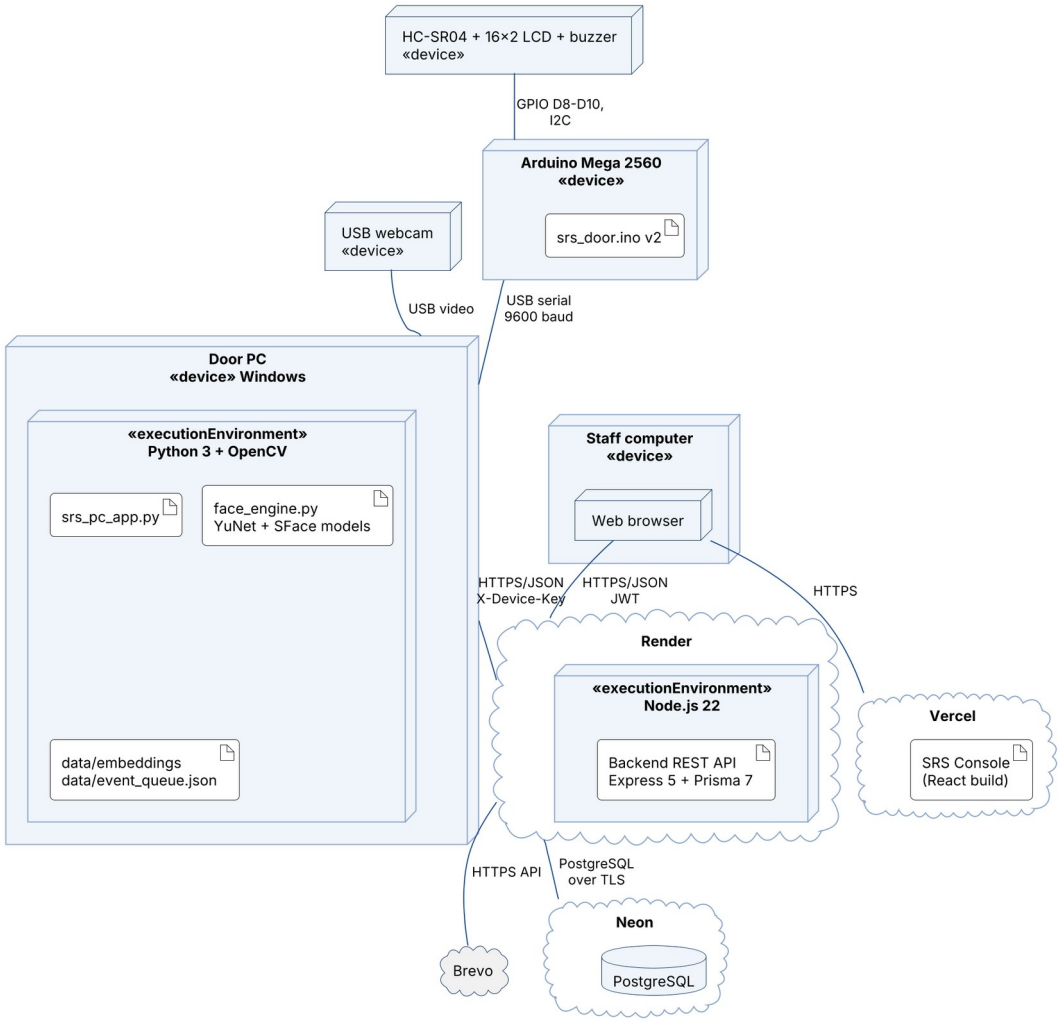
Next steps

- On-site FAR/FRR test with 30+ volunteers
- Liveness check against printed photos
- Move the door program to a Raspberry Pi 5
- Import students from the university system

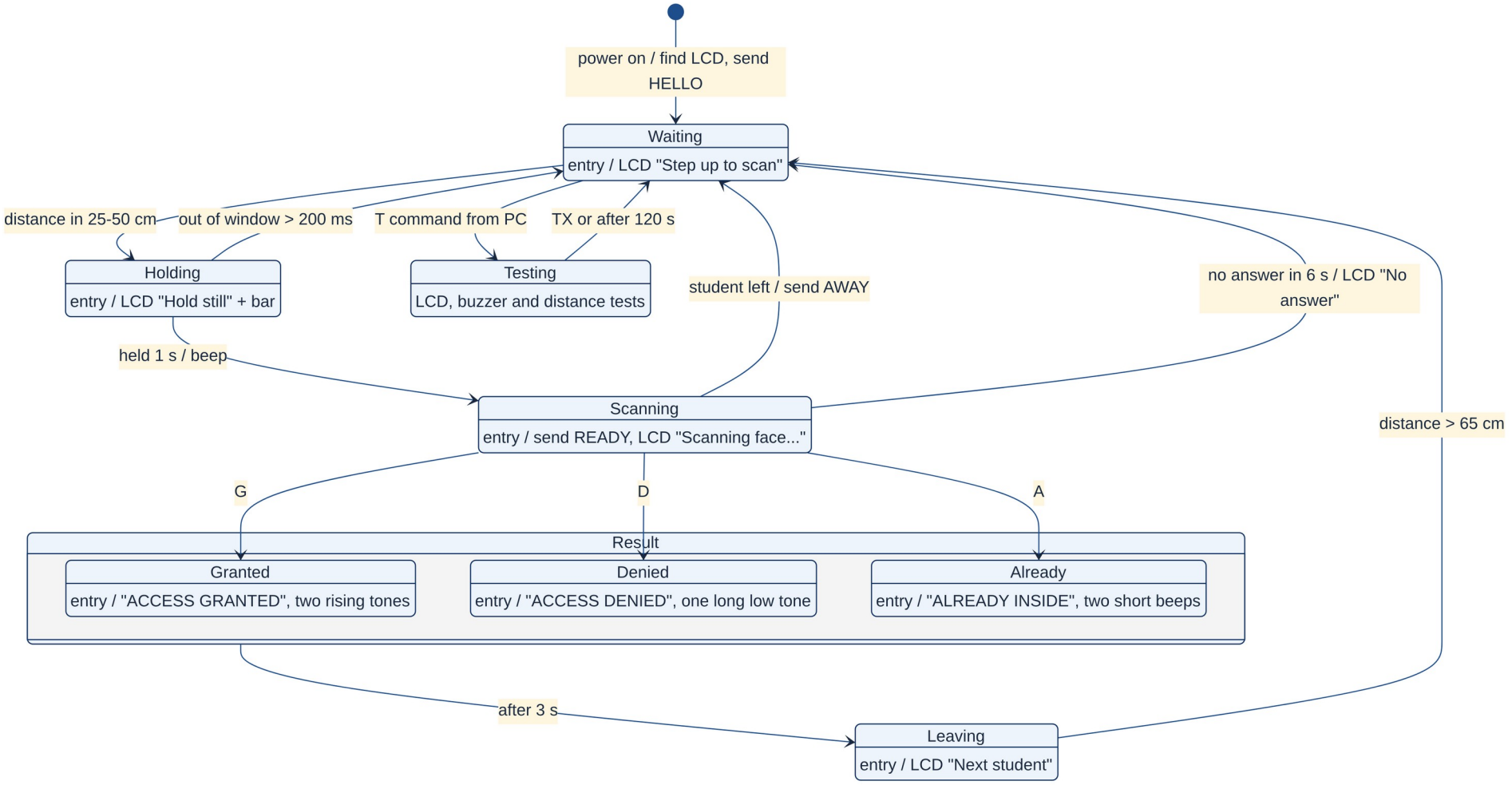


Thank you – live demonstration and questions

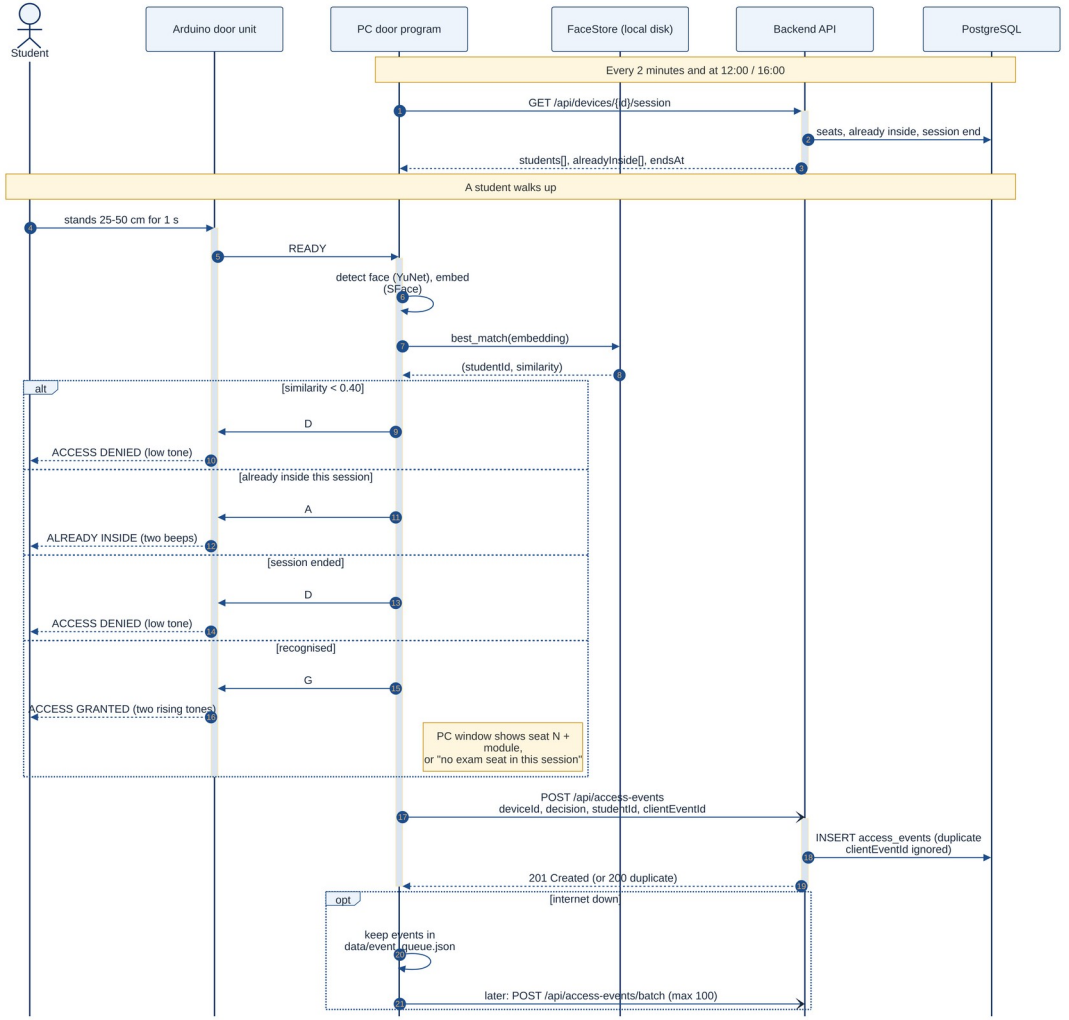
Backup: UML deployment diagram



Backup: UML state machine of the door unit



Backup: UML sequence – one access verification



Backup: UML sequence – booking, e-mails, sign-out

